

Activity

Part 2: The Case — “Dr. Reyes’s Grassland Study” (20 min)

Setup (3 min)

You are a peer reviewer for the journal Ecological Applications. You’ve just received the following methods section. Your job: find everything that’s wrong — and explain why it matters.

The Manuscript Excerpt — Read Carefully

Title: *Nitrogen addition increases aboveground biomass in semi-arid grasslands*

Methods (excerpt):

We established 12 plots across 4 ranches in eastern Colorado. Each ranch received one of three nitrogen treatments: control (0 kg N/ha), low (25 kg N/ha), and high (50 kg N/ha). Biomass was harvested from each plot at three time points: May, July, and September of the same year.

To assess treatment effects, we calculated mean biomass per treatment per time point and ran a one-way ANOVA (treatment as the fixed effect) on the September harvest only, as this represented “peak biomass.” We report p-values and conclude that high nitrogen significantly increased biomass ($p = 0.03$). All plots were located within 5 km of one another on each ranch.

We selected ANOVA because it is the standard method in grassland ecology and because our dataset was small. We acknowledge that data were not fully independent but considered this a minor concern.

Individual choice of reject, revisions or accept.

Part 3: Case Diagnosis — Small Group Work (20 min)

Work in groups of 3–4. Each group gets **one diagnostic role** you are responsible for that angle, but your findings feed the whole class.

Role Assignments

Role	Your Diagnostic Focus
Triangle Auditors	Does the method match the question, data type, and structure? What would the correct model be?
Framework Detectives	What philosophical framework is Dr. Reyes implicitly using? Where does it break down? What would a Bayesian alternative look like?
Complexity Judges	Was dropping July and May data defensible? What is lost? Is the model too simple? Could it have been simpler in a <i>different</i> direction?
Reproducibility Auditors	What would you need to reproduce this analysis? What is missing? List <i>specific</i> items.

Group Task

1. Read the excerpt again as a group
2. Prepare a **2-minute verbal finding**: one clear diagnosis + one concrete fix